

Introduction to predicate logic.

- Propositions

Ex: $\left. \begin{array}{l} 2 \text{ is even} \\ 2 \text{ is odd} \end{array} \right\} \text{propositions.}$

$n \text{ is even} : \text{not a proposition}$

- Predicates: predicate logic is more general type of logic that allows us to write function like logical expressions called **predicates**

- Informally, predicate is a property that a particular entity may or may not have.

- A proposition with blanks

ex: The integer is a prime

* once the blanks of a predicate is filled, then it becomes a proposition.

The integer 57 is not a prime

Def A predicate p is a function that assigns value True or False to each element of a Set U .

(predicate P is a function $P: U \rightarrow \{\text{True}, \text{False}\}$)

Ex 1:

$$\text{IsPrime}(n) = \begin{cases} \text{True} & , \text{if } n \text{ is a prime} \\ \text{False} & \text{otherwise} \end{cases}$$

$$U = \mathbb{Z}$$

$\text{isPrime}(n)$ $\leftarrow$ not a proposition

$\text{isPrime}(2)$

$\text{isPrime}(6)$

$$2. \text{IsEven}(n) = \begin{cases} \text{True} & , \text{If } n \text{ is even} \\ \text{False} & , \text{otherwise} \end{cases}$$

$$U = \mathbb{Z}$$

$$3. \text{ Is Rational } (x) = \begin{cases} \text{True} & , \quad x \in \mathbb{Q} \\ \text{False} & , \quad x \notin \mathbb{Q} \end{cases}$$

How can you make a predicate a proposition?

1. Apply the predicate to a specific element of the universe.

Suppose we have $\text{isEven}(x)$,
then let $x = 7$

$\text{isEven}(7)$ is a proposition

let $x = 6$

$\text{isEven}(6)$ is a proposition.

Q: for all integers n $\text{isEven}(n)$ is true.

This is a proposition and truth value of it is false.

There exists some integer n such that $\text{isEven}(n)$ is true.

This is a proposition and its truth value is true.

2. Using quantifiers.

a) Universal quantifier.

$\forall$ "for all"

$[\forall n \in \mathbb{Z} : \text{isEven}(n)]$

$[\forall n \in \mathbb{Z} : \neg \text{isEven}(n)]$

$$\forall x \in U: P(x)$$

"for all x in U , $P(x)$ is true"

$[\forall x \in U: P(x)]$ to Alternate way
view universal
quantifier

$$U = \{x_1, x_2, x_3, \dots, x_n\}$$

$$P(x_1) \wedge P(x_2) \wedge P(x_3) \wedge \dots \wedge P(x_n)$$

$$[\forall x \in U: P(x)] = P(x_1) \wedge P(x_2) \wedge P(x_3) \wedge \dots \wedge P(x_n)$$

Remember the theorem

Given $n \in \mathbb{Z}$

If n^2 is even, then n is even.

$$[\forall n \in \mathbb{Z}: \text{If } n^2 \text{ is even, then } n \text{ is even}]$$

$[\forall n \in \mathbb{Z}: n^2 \text{ is even} \Rightarrow n \text{ is even}]$

$G(n): n^2 \text{ is even} \Rightarrow n \text{ is even}$

$[\forall n \in \mathbb{Z}: G(n)]$

$\equiv$

$G(1) \wedge G(2) \wedge G(3) \wedge \dots \wedge$

$[\forall n \in \mathbb{Z}: (1^2 \text{ is even} \Rightarrow 1 \text{ is even}) \wedge$
 $(2^2 \text{ is even} \Rightarrow 2 \text{ is even}) \wedge$
 $\vdots$
 $(\dots)]$

In natural languages, sometimes we omit the explicit mention of quantifiers.

From here onwards, we will use quantifiers along with predicates to write logically sound propositions.

b) Existential quantifier.

$\exists$ "there exists"

$$[\exists x \in U: P(x)]$$

"There exists an element x in U , such that $P(x)$ is true.

$$[\exists x \in \mathbb{Z}: \text{isEven}(x)] \quad \text{True proposition}$$

$$[\exists x \in \mathbb{Z} : \neg \text{isEven}(x)]$$

we can view this proposition in an alternate form.

$$U = \{x_1, x_2, x_3, \dots, x_n\}$$

$$P: U \longrightarrow \{\text{True}, \text{false}\}$$

$$[\exists x \in U: P(x)]$$

$$P(x_1) \vee P(x_2) \vee P(x_3) \vee P(x_4) \vee \dots \vee P(x_n)$$

Exercises

1. for all integers n , $2n$ is even

$$[\forall n \in \mathbb{Z}: \text{is Even}(2n)] \quad \text{True}$$

2. for all integers n , $2n+1$ is odd

$$[\forall n \in \mathbb{Z}: \neg \text{is Even}(2n+1)]$$

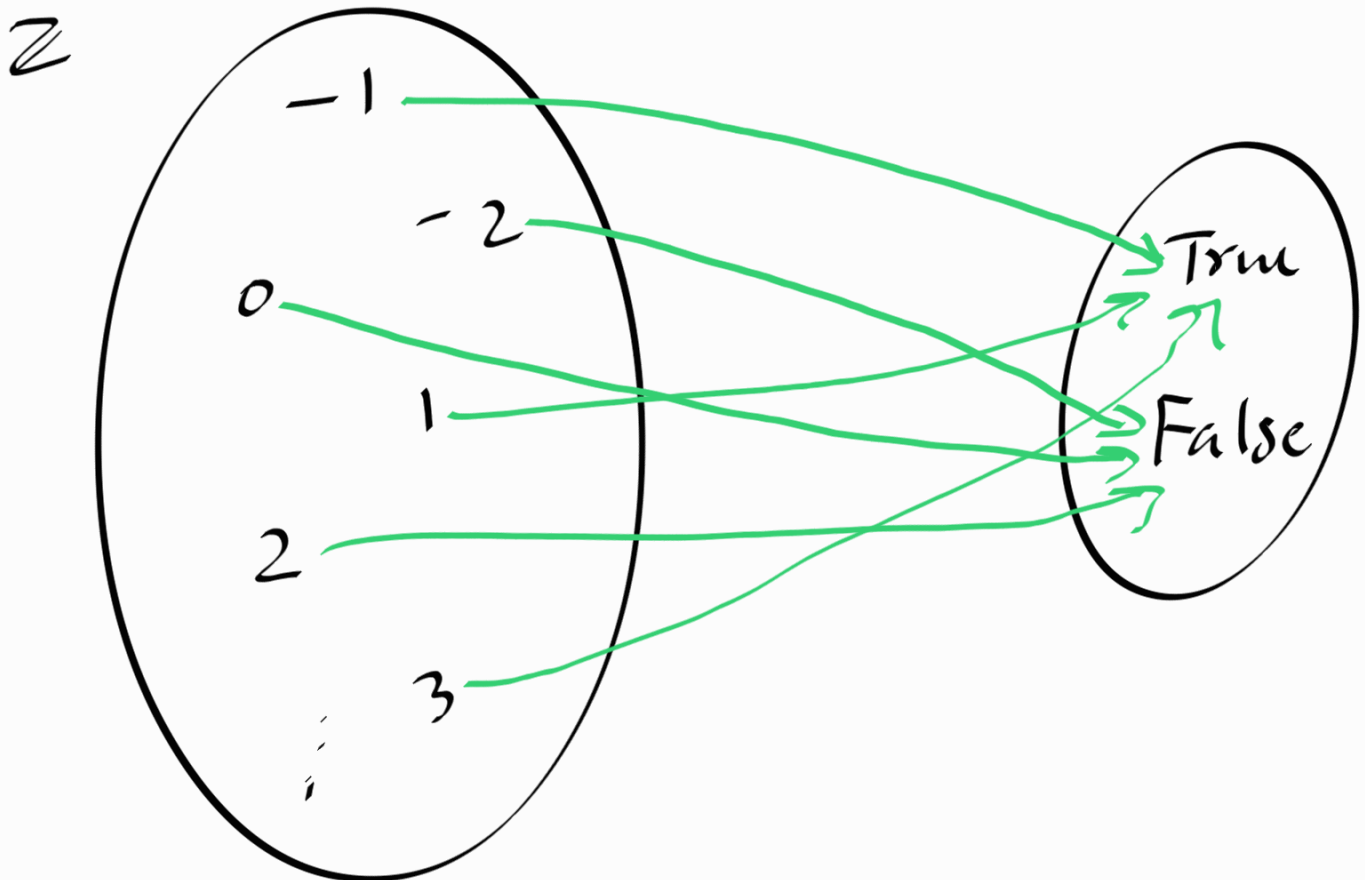
$$[\forall n \in \mathbb{Z}: \text{is Odd}(2n+1)] \quad \left. \vphantom{[\forall n \in \mathbb{Z}: \text{is Odd}(2n+1)]} \right\} \text{True}$$

3. There exists real numbers x, y such that xy is rational and not (both x and y are rational)

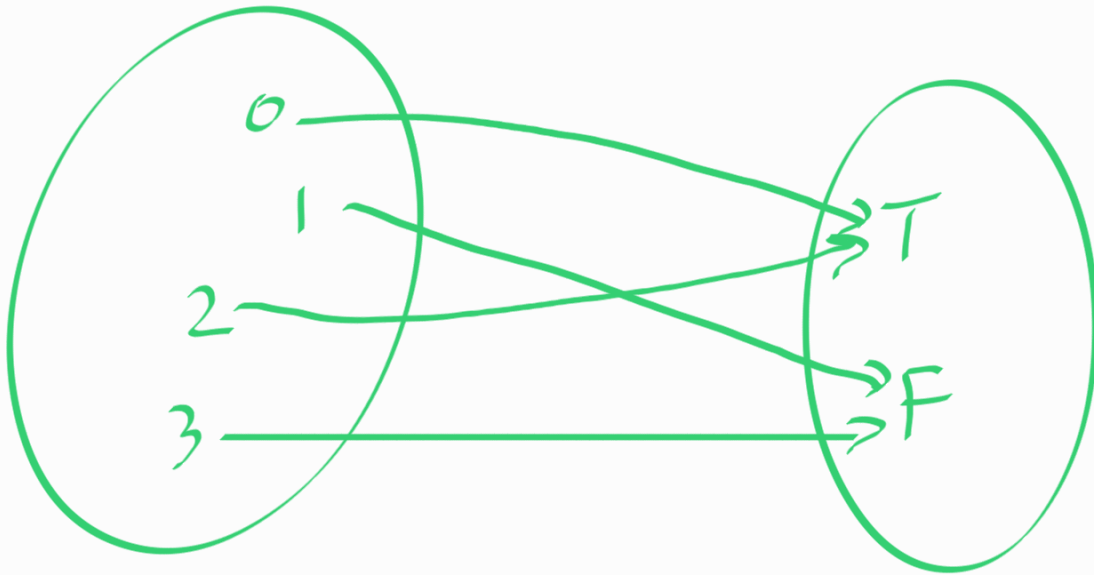
$$[\exists x, y \in \mathbb{R}; xy \in \mathbb{Q} \wedge \neg(x \in \mathbb{Q} \wedge y \in \mathbb{Q})]$$

Ex: isOdd: $\mathbb{Z} \rightarrow \{T, F\}$

$$\text{is odd}(x) = \begin{cases} \text{True, } x \text{ is odd} \\ \text{False, otherwise} \end{cases}$$



$\text{isEven} : \{0, 1, 2, 3\} \rightarrow \{T, F\}$



Predicate itself is not a proposition.

Def Theorems in predicate logic.

A fully quantified expression of predicate logic is a theorem if and only if it is true for every possible meaning of its predicates

Theorem 3.39

$$P: S \longrightarrow \{T, F\}$$

$$[\forall x \in S : P(x) \vee \neg P(x)]$$

$$S = \{x_1, x_2, x_3, \dots, x_n\}$$

$$[\forall x \in S : P(x) \vee \neg P(x)] \equiv \underbrace{(P(x_1) \vee \neg P(x_1))}_T \wedge \underbrace{(P(x_2) \vee \neg P(x_2))}_T \wedge \dots$$

$$\vdots$$

$$(P(x_n) \wedge \neg P(x_n))$$

T

Theorem 3.40 $\equiv$ True.

$$[\forall x \in S : P(x)] \vee [\forall x \in S : \neg P(x)]$$

This is not true

Suppose isEven is a predicate defined as follows.

$$\text{isEven} : \{0, 1, 2, 3\} \rightarrow \{T, F\}$$

$$[\forall x \in \{0, 1, 2, 3\} : \text{isEven}(x)] \vee [\forall x \in \{0, 1, 2, 3\} : \neg \text{isEven}(x)]$$

$$\equiv$$

$$[\text{isEven}(0) \wedge \text{isEven}(1) \wedge \text{isEven}(2) \wedge \text{isEven}(3)]$$

$$\vee$$

$$[\neg \text{isEven}(0) \wedge \neg \text{isEven}(1) \wedge \neg \text{isEven}(2) \wedge \neg \text{isEven}(3)]$$

$$F \vee F \equiv F$$

This is a disproof by counter example.

Theorem 3.41

$$\neg [\forall x \in S: P(x)] \equiv [\exists x \in S: \neg P(x)]$$

$$\neg (P(x_1) \wedge P(x_2) \wedge \dots \wedge P(x_n)) \equiv$$

$$\neg P(x_1) \vee \neg P(x_2) \vee \neg P(x_3) \vee \dots \vee \neg P(x_n)$$

$$\equiv [\exists x \in S: \neg P(x)]$$

Theorem 3.42

$$\neg [\exists x \in S: P(x)] \equiv [\forall x \in S: \neg P(x)]$$

Theorem 3.43

$$[\forall x \in \phi : p(x)]$$

All unicorns are blue